

Aayushi Choksi

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Summary

Full-Stack Developer with 2+ years of experience building scalable, secure applications with expertise in cloud infrastructure, DevOps, and performance optimization. Proficient in React, Node.js, Spring Boot, and deploying CI/CD pipelines and microservices on AWS.

Work Experience

Software Engineer

May 2025 – Present

Humanitarians AI

- Built a full-stack **LLM**-powered natural language to SQL interface using **Claude** via **Amazon Bedrock**, **LangChain**, and **Streamlit**, enabling users to query structured data in **Amazon RDS** (PostgreSQL) without writing SQL.
- Integrated Claude with a **Python** backend (**boto3**, **psycopg2**) to dynamically generate and execute SQL queries, achieving 90% accuracy across 50+ diverse query intents and delivering secure, real-time business insights.

Research Software Engineer

Sept 2024 – Present

Northeastern University

- Engineered full-stack traffic monitoring system using **Flask** and **PostgreSQL**, implementing scalable Python **ETL** pipelines that processed 500K+ freeway data points with 92% anomaly detection accuracy.
- Built interactive dashboards using **React**, **D3.js**, and **TypeScript**, integrating real-time analytics for traffic patterns, vehicle movement, and congestion analysis, with features like pack formation tracking, density analysis, and dynamic charting.
- Developed secure, high-performance **RESTful APIs** to connect backend analytics with frontend interfaces, implementing token-based authentication and optimizing data delivery for smooth user interaction.

Software Development Engineer

Feb 2022 – Aug 2023

Promount Technologies

- Engineered distributed microservices in **Java** using **Spring Boot**, **Kafka**, and **ActiveMQ**; implemented **Redis** caching for low-latency response handling, improving throughput and API performance by 40%.
- Built an **Angular-TypeScript** dashboard with **NgRx**, improving role-based access control and UI responsiveness.
- Implemented **OAuth2** with JWT-based authentication, enforced **IAM policies** and API Gateway security, and developed unit tests with **JUnit**, reducing unauthorized access by 35% and ensuring high test coverage.
- Containerized and deployed with **Docker** and **Kubernetes** (EKS), handling 5x traffic spikes, reducing costs by 30%.
- Designed **CI/CD** pipelines with **Jenkins**, **GitHub Actions**, and **Terraform**, integrating **Kibana** and **CloudWatch** for monitoring. Resolved 70+ production issues via log tracing, reducing bugs by 20% and accelerating release cycles by 40%.

Software Engineer

May 2021 – Dec 2021

Bhaskaracharya National Institute for Space Applications and Geo-informatics

- Developed a secure e-commerce platform in **Python**, **Django**, and **JavaScript**, integrating **GraphQL** for flexible API queries and **NoSQL** (**MongoDB**) for scalability.
- Implemented MFA and AES encryption to secure user authentication, reducing unauthorized access by 40%.
- Integrated a **web application firewall (WAF)** to mitigate SQL injection and XSS attacks, ensuring 99% data integrity.
- Collaborated with a team of 8+ developers, designers, and QA testers in an **Agile** environment using **Scrum** and **Git** for version control to drive iterative improvements and continuous delivery.

Projects

IAC-Powered Cloud Application Platform

- Designed and deployed a secure backend with **Node.js** and **Sequelize** for **MySQL**, integrating **CI/CD** pipelines using **GitHub Actions** and **Packer** to automate testing and AMI creation, reducing build errors by 40%.
- Used Terraform to provision scalable **AWS** infrastructure (EC2, RDS, Route53, IAM, Load Balancer, Auto Scaling, CloudWatch). Built a **serverless** authentication workflow in **Golang** using Lambda, SQS, and Mailgun for secure email verification.

Husky Connect

- Built a modular **React.js** UI with secure JWT-based session control using Node.js and **Express**, enabling personalized dashboards and real-time feeds. Integrated **Socket.io** to develop a real-time chat feature that boosted user engagement by 25%.
- Implemented **GraphQL** with **MongoDB** and DataLoader for efficient data handling, and deployed the application on AWS EC2 with Docker Compose to support auto-scaling and load balancing.

Education

Northeastern University (GPA: 3.8/4.0)

Boston, MA

Master of Science in Information Systems

Sept 2023 – May 2025

Roles: Graduate Teaching Assistant for Data Structures & Algorithms course.

Ganpat University (GPA: 3.62/4.0)

Gujarat, IN

B.Tech Computer Science and Engineering with specialization in Cloud Computing

Jun 2019 – May 2023

Technical Skills

- Programming:** Java, Python, Go, C, C++, JavaScript, Typescript, HCL, PHP, HTML5, CSS3, Shell scripting, Rust
- Frameworks:** React, Node.js, Express.js, GraphQL, Spring Boot, Redux, MVC, J2EE, Kafka, Sequelize, Hibernate
- Tools & Databases:** Git, MongoDB, MySQL, Postman, Terraform, Selenium, Jenkins, Docker, Kubernetes, Redis
- Cloud Tech:** AWS (EC2, S3, SQS, SNS, Lambda), GCP (VPC, DNS), Microsoft Azure, DigitalOcean